DA&MTSS: An End-to-End Remote Sensing Image Domain Adaptive Semantic Segmentation Framework Combining Data Augmentation and Mobile Threshold Self-Supervision

Dong Chen, Yuebin Wang, *Member, IEEE*, and Liqiang Zhang

***Abstract*— The application of deep learning-based semantic segmentation in remote sensing (RS) images has achieved considerable success. However, many supervised methods still heavily rely on a large amount of labeled data, requiring time-consuming and labor-intensive manual annotations. Besides, networks trained on labeled source domain data often perform poorly in inference tasks with target domain data due to the domain shift phenomenon. To address these challenges, we construct a novel end-to-end unsupervised domain adaptation (UDA) framework, named data augmentation and mobile threshold self-supervision (DA&MTSS), which integrates data augmentation with self-supervision. Specifically, we analyze the common factors that cause domain shifts in RS images and adopt different data augmentation techniques to attenuate the domain shift and enhance the generalization ability and robustness of the network in cross-domain inference. In the self-supervision phase, we design a new sample-based mobile threshold method to dynamically control the thresholds of both dominant and long-tail classes during training and generate stable pseudo-labels. Therefore, our method eliminates the need for additional training or expert knowledge and achieves the co-evolution of network parameters and pseudo-label quality in the training process. The results of comprehensive experiments on five tasks across the ISPRS Vaihingen, Potsdam, and LoveDA datasets demonstrate that this method consistently achieves higher mIoU scores, showcasing the performance advantage of DA&MTSS in UDA for the semantic segmentation of RS image.**

***Index Terms*— Data augmentation, reliable pseudo-labels, remote sensing (RS) image, self-supervision, self-training, unsupervised domain adaptation (UDA) semantic segmentation.**

Manuscript received 22 January 2024; revised 27 April 2024 and 28 June 2024; accepted 19 July 2024. Date of publication 23 July 2024; date of current version 1 August 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 41925006, Grant 42171388, and Grant 41801241; in part by the Fundamental Research Funds for the Central Universities under Grant 292018029 and Grant 375201906; in part by the Key Research and Development Projects of Shanxi Province under Grant 201903D121142; and in part by Guizhou Science and Technology Plan Project under Grant Qian Ke He Zhi Cheng [2020] 4Y022. *(Corresponding author: Yuebin Wang.)*

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Digital Object Identifier 10.1109/TGRS.2024.3432711

I. Introduction

IN RECENT years, remote sensing (RS) technology has rapidly advanced, which brings a wealth of RS images, highlighting the increasing significance of automated interpretation techniques. The full utilization of such advanced technologies as deep learning to achieve the automatic analysis and understanding of high-resolution RS images can provide precise and efficient information support for environmental monitoring, resource management, disaster response, and other fields. As a key technology for automatic interpretation, semantic segmentation allows for a more accurate understanding and utilization of information within RS images. Over the past few years, the convolutional neural network (CNN) has established its superiority in feature representation in the field of computer vision (CV) [1], [2], [3], [4], and the fully convolutional network (FCN) [5], as a classical framework, has successfully been applied to semantic segmentation. It has been verified in numerous studies that RS image semantic segmentation has demonstrated its notable advantages over conventional methods.

However, the dependence of RS image semantic segmentation on a large amount of labeled data restricts its applicability due to the demand for time-consuming and labor-intensive manual pixel-level annotations. This task becomes even more challenging for RS images due to their high resolution and rich land details. Although many semantic segmentation datasets have dense annotations, the domain shift phenomenon between different datasets often leads to poor performance of networks trained on the source domain in the direct inference of the target domain. The domain shift is a common phenomenon in RS images, as images provided by different satellites or sensors are affected by various factors, such as time, angle, and atmospheric environment. Specifically, these differences are mainly reflected in different spatial resolutions, changing lighting conditions, shadows of buildings or trees, and different data acquisition sensors.

In the field of CV, unsupervised domain adaptation (UDA) has been proposed to resolve performance degradation in cross-domain inference due to domain shifts. The basic principle of UDA involves the alignment of the source domain with the target domain, which can be mainly divided into

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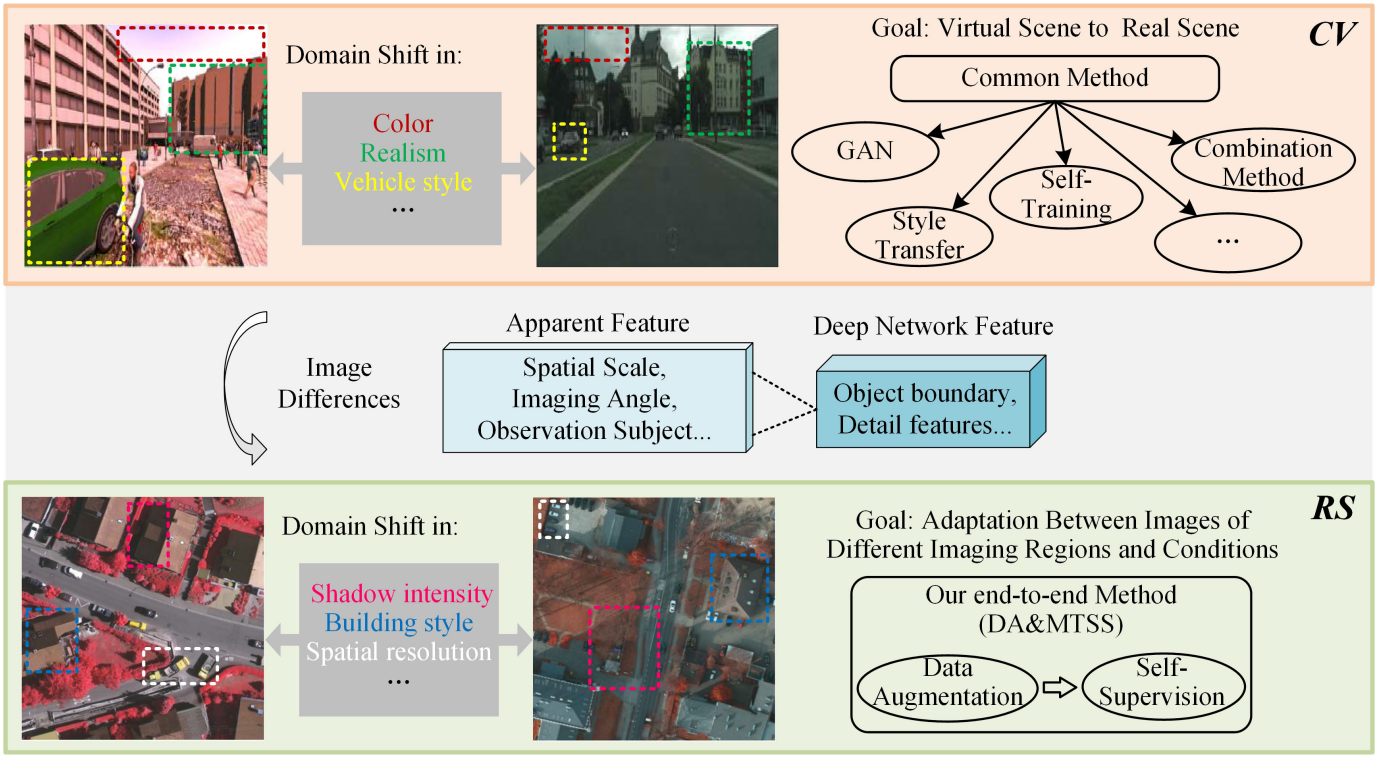


Fig. 1. Differences in UDA semantic segmentation between CV and RS: the causes of domain shift, the differences between natural images and RS images, and the differences in goal tasks.

feature-level, image-level, and output-level adaptation [6]. UDA is typically related to adversarial learning [7], style transfer [8], and other methods. Many studies have achieved significant success in UDA semantic segmentation tasks, but they are primarily designed for natural images [9]. With the rapid advancement in computer graphics and platforms, an infinite number of realistic virtual scenes and corresponding annotations can be generated. This advancement presents a unique opportunity for reducing the domain shift between synthetic virtual scenes and real scenes in CV. On that basis, the model deployed on real data can perform well, and manual annotations on real scenes can be avoided. In contrast to the domain shift in RS images, that in natural images mainly is manifested as appearance differences, such as realism, lighting, color, texture, and class differences in vehicle style, pedestrian clothing, and building style. RS images are characterized by larger spatial scales, unique imaging angles, diverse observation subjects, complex scenes, and tight spatial layouts. The deep network features in RS images have a lower resolution, which may result in lost object boundaries and detail features. Additionally, RS images contain a wider geographic range, and more abundant semantic context information needs to be considered for deep features. As shown in Fig. 1, examples visually illustrate the domain shift differences, image differences, and goal differences faced by the CV and RS in UDA tasks. Due to these unique challenges, the aforementioned methods often perform poorly in UDA tasks for RS images. Currently, many methods have been adopted to address UDA tasks in RS by drawing inspiration from relevant methods in CV. However, it is often required for these methods to add cumbersome preprocessing or post-processing modules or conduct multiple rounds of training, which may be time-consuming and require continuous expert intervention. Further, the underlying cause of domain shifts may be neglected in these methods. Therefore, it is necessary to develop novel and effective UDA methods specifically for RS images, which may address these challenges and improve performance without requiring extensive preprocessing or expert intervention.

To address these challenges, the end-to-end data augmentation and mobile threshold self-supervision (DA&MTSS) framework is proposed (refer to Fig. 2), which only requires a single training round. Initially, data augmentation techniques are employed to simulate common domain shift phenomena observed in RS images. This enhances the network's generalization and robustness for cross-domain inference. In the self-supervision phase, a duplicate of the baseline network serves as the student model. Momentum-based methods are applied to the gradual update of the network, thus generating stable pseudo-labels. To tackle the prevalent class imbalance in RS images, thresholds for pseudo-label supervision are dynamically computed using a sample-based mobile threshold method. During the iterative process of updating parameters in network training, the threshold of pseudo-labels is also iteratively updated. This eliminates the demand for multiple rounds of training and expert intervention, thus realizing end-to-end training, which can be called "co-evolved." The contributions can be summarized as follows.

1) A novel sample-based mobile threshold method is designed, which can simultaneously control the

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thresholds of both dominant and long-tail classes and generate stable pseudo-labels.

2) An end-to-end DA&MTSS framework is proposed based on the mobile threshold method, which eliminates the need for additional training or expert knowledge, realizing the co-evolution of network parameters and pseudo-label quality in the training process.
3) The reasons for inter-domain differences in RS images are analyzed for the first time, and various data augmentation strategies are specifically employed to reduce domain shifts.
4) This method outperforms baselines and other advanced techniques, achieving optimal results. Comprehensive experiments on five tasks across three public datasets have demonstrated the significant advantages of DA&MTSS in UDA semantic segmentation tasks for RS images.

II. Related Works

This section offers a summary of UDA semantic segmentation in both CV and RS domains, as well as related self-training and pseudo-labeling methods related to the proposed method.

A. Exploration in CV

The basic principle of UDA involves the alignment of the source domain with the target domain, which can be achieved through feature-level, image-level, and output-level adaptive methods [6]. In feature-level adaptation, maximum mean discrepancy [10] and the incorporation of a discriminator [11] through adversarial learning [7] have been successful in the UDA image classification field. These methods have been applied to UDA semantic segmentation tasks. Hoffman et al. [12] combined adversarial learning and constrained weak learning for global and class-specific alignment, respectively. Chen et al. [13] further extracted prior knowledge from static images and achieved improved performance in road scenes. Hong et al. [14] relied on the conditional generator to transform the source features into class-specific target features and used a multilayer perceptron as the domain discriminator. Zhang et al. [15] utilized three unlabeled constraints as model regularizers to improve the alignment. However, if there is a significant difference in the feature distribution between two domains, feature-level adaptation cannot achieve the alignment effectively, leading to a noticeable decline in its performance. Additionally, such methods also encounter such problems as information loss, overfitting, and difficulty in handling subtle domain differences.

In image-level adaptation, the style transfer [8] method is commonly used to transform the appearance of the domain for domain adaptation. Many scholars (Liu and Tuzel [16], Bousmalis et al. [17], Taigman et al. [18], Chang et al. [19], Chen et al. [20], and Yang et al. [21]) have demonstrated the efficacy of style transfer in UDA semantic segmentation tasks, following its success in image classification. Hoffman et al. [22] selected the cycle consistency loss and semantic consistency loss as the most commonly used regularization methods for style transfer. Inspired by the study of Hoffman et al. [22], Toldo et al. [23] employed CycleGAN [24] to align the synthetic domain with the real domain. Yang et al. [25] reconstructed images from predicted labels to reduce the bias introduced by image translation.

Output-level adaptation involves the direct addition of adversarial adaptation to the output space to avoid the instability and unreliability of high-dimensional feature alignment. This method was first implemented by Tsai et al. [26] to construct a clustered space and jointly train the segmentation network and classifier. Vu et al. [27] derived the weighted self-information mapping (wSIM), combining the entropy loss and adversarial loss for training. Pan et al. [28] considered both inter-domain differences and intra-domain differences, and they minimized both differences through adversarial learning and self-supervision, respectively. Wang et al. [29] effectively exploited low-confidence predictions by taking account of the spurious label noise caused by domain shifts.

B. Exploration in RS

Although many methods have achieved impressive results in UDA semantic segmentation tasks for natural images, they often perform poorly in the semantic segmentation of RS images, which exhibit significant differences in image contents and domain shifts. In recent years, various methods have been introduced to specifically address the challenges posed by domain shifts in RS images, with the aim of avoiding the need for manual annotations. Wittich and Rottensteiner [30] designed an appearance-adaptive method with semantic consistency by jointly training an appearance-adaptation network and classifier. Zhang et al. [31] proposed a staged framework (called RoadDA) to resolve the domain shift in road segmentation. Li et al. [32] minimized the domain shift using multiple weak supervision constraints in the objective function. Shen et al. [33] divided the predicted output into easy and hard segments, and they utilized self-supervision to enhance the performance of hard segments. Liu et al. [34] fine-tuned the model using knowledge distillation, improving its performance in three stages. Chen et al. [35] designed an invariant feature memory module based on class attention to augment the representation of pseudo-invariant features. Li et al. [32] optimized the model with multiple weak supervision constraints for cross-domain inference in RS. Huang et al. [36] proposed a semi-supervised method combining unsupervised alignment and supervised alignment through bidirectional sample alignment, thus achieving optimal performance in scene classification. Chen et al. [37] reduced domain shifts using scene simulation and geometry simulation by introducing the S&GDA framework. Li et al. [38] designed the semantic preservation module (SPGAN) and class distribution alignment module (CDA), which were coupled into the SPGAN-DA framework suitable for UDA tasks.

C. Self-Training and Pseudo-Label

Self-training is a type of method that can be employed to improve the model iteratively using high-confidence class predictions as pseudo-labels. To select reliable pseudo-labels,

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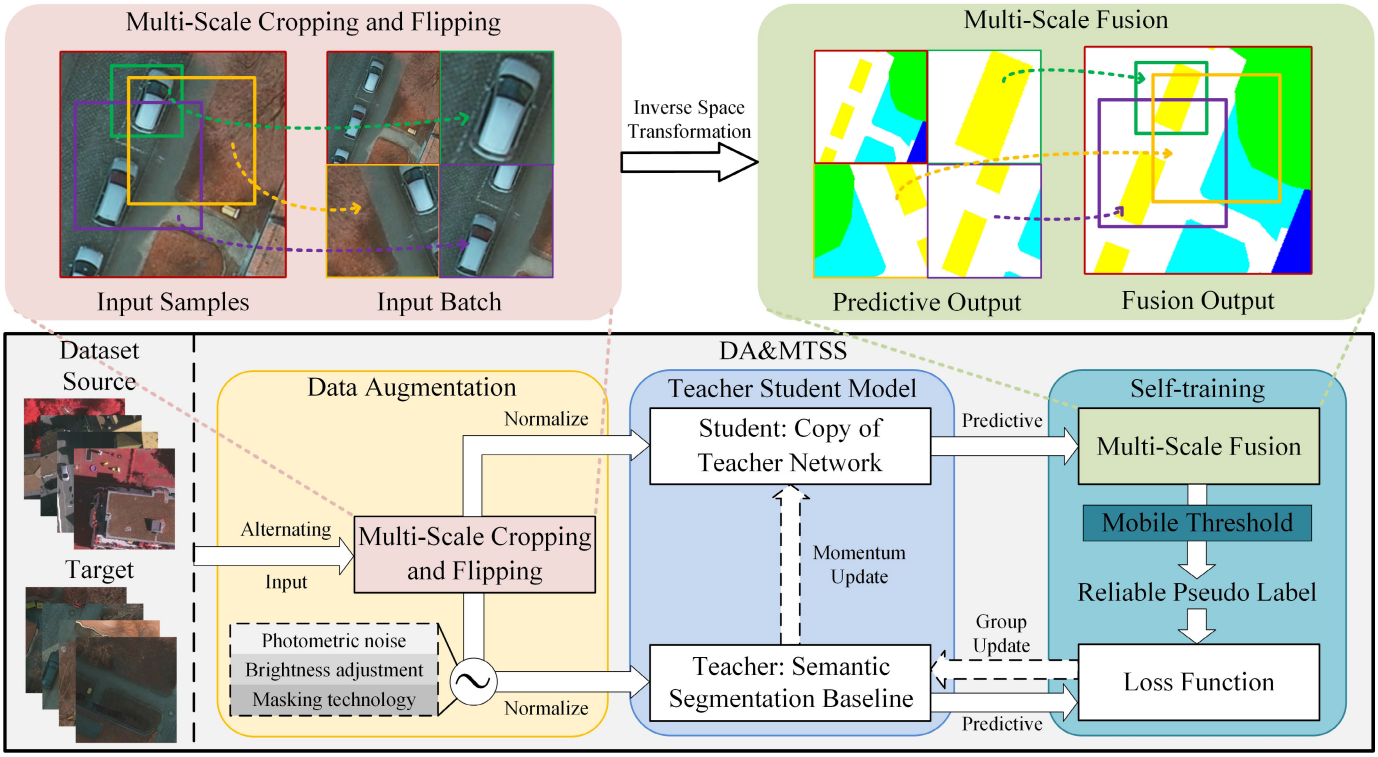


Fig. 2. End-to-end framework of DA&MTSS. Specifically, various data augmentation techniques weaken the shift between domains and improve the network's generalization ability and robustness during cross-domain inference. Secondly, self-supervised training is performed using co-evolving pseudo-labels to avoid the demand for multiple rounds of training and expert intervention.

Li et al. [39] and Wang et al. [40] extrapolated the confidence of pseudo-labels from the maximum class probability output by the model. Shen et al. [41] simultaneously considered the confidence of both the prediction output and the discriminator to identify higher-quality pseudo-labels. Zheng and Yang [42] introduced memory regularization in vivo to regularize the training process. Zheng and Yang [43] corrected pseudo-label learning by estimating the prediction uncertainty. Tranheden et al. [44] proposed the DACS framework, in which both labeled and pseudo-labeled data were combined in training samples. Guo et al. [45] enhanced the reliability of pseudo-labels through meta-correction. Zou et al. [46] proposed the UAPL framework, demonstrating its effectiveness in cross-scene target detection tasks on high-resolution synthetic aperture radar (SAR) images.

III. Methodology

A. Overall Framework

Fig. 2 illustrates the two main components of DA&MTSS: data augmentation and self-supervision (teacher–student model and self-training). Firstly, the self-supervised domain adaptation process is initiated by the alternation between source and target domain images. This process involves the application of random multiscale cropping and horizontal flipping. Then, the input data for the semantic segmentation baseline network undergoes additional augmentation with random photometric noise, brightness adjustment, and masking techniques. To adapt to the target domain, the baseline network is duplicated as a student network, and its parameters are gradually updated through momentum gradient descent. Once the student network predicts the target domain data, the multiscale fusion of cropped images is performed through inverse spatial transformations. The average prediction is obtained as the semantic mask. The threshold for each class is dynamically identified as a pseudo-label using a sample-based mobile threshold method. High-confidence semantic masks are filtered out to generate pseudo-labels. The parameter update of the segmentation network is guided by real labels from the source domain and pseudo-labels from the target domain. Meanwhile, the slowly updated student network contributes to the production of more stable pseudo-labels. In Sections III-B–III-D, we will provide a detailed exploration of the individual sub-modules through our method.

B. Data Augmentation for Reducing Domain Shifts

RS images captured by different satellites or sensors may be affected by various factors, such as time, angle, and atmospheric environment, thus resulting in significant differences between RS images from different domains. Specifically, these differences are mainly reflected in different spatial resolutions, changing lighting conditions, shadows of buildings or trees, and different data acquisition sensors. In this study, different data augmentation techniques are employed to simulate inter-domain gaps and input them into the network for training. These efforts may weaken the impact of inter-domain gaps on the network and improve the generalization ability and robustness of the network.

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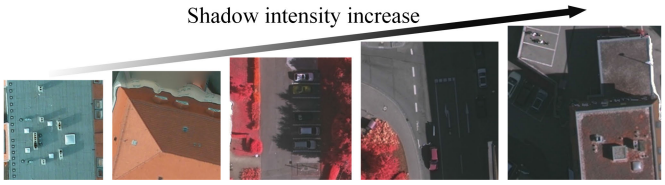


Fig. 3. Shadows of different intensities resulting from different causes. From left to right, they are object shadows due to the solar altitude angle, shadows caused by sloping roofs, shadows caused by tree obstruction, shadows caused by building obstruction, and shadows caused by a group of buildings.

1) Multiscale Cropping and Flipping: It involves randomly cropping and horizontally flipping the input image with a set threshold, as well as the restoration of the input image to the original image size through bilinear interpolation. In this process, detailed images of different resolutions can be generated to adapt to the impact of different spatial resolutions on the segmentation network, as shown in the upper left of Fig. 2.

2) Photometric Noise: It involves the random application of color jitter and Gaussian filtering to the image to encourage the segmentation network's invariance to photometric perturbations.

3) Brightness Adjustment and Masking: Shadows generated by occlusion and lighting angles usually appear as black or gray areas. In this study, the random brightness adjustment of local areas is used to simulate simple shadow types. However, the shadows with different strengths are generated by different lighting directions, and the objects that cause occlusion are also different, as shown in Fig. 3. Based on that, masking techniques can be utilized to randomly occlude local areas of the image to simulate more complex shadow types and weaken the impact of shadow differences between different domains on the segmentation network.

C. Reliable Pseudo-Labeling for Self-Supervised Training

1) Teacher–Student Network: In this study, a semantic segmentation baseline is used as the teacher network, with the student network being an identical copy of the teacher network. After the teacher network updates its gradient through backward propagation, the parameters of the student network are slowly updated using momentum gradient updates. The photometric noise, brightness adjustment, and masking used in data augmentation are only input into the teacher model, and "clean" inputs are provided for the student model. This improves the robustness of the segmentation network and provides a stable objective for self-supervised training of the segmentation network.

2) Multiscale Fusion: The output mask of the student network is re-projected back to the cropped size and position. To avoid incomplete coverage of the cropped area, the original image is forward propagated through the student network. For each pixel, the predicted softmax values are averaged in the overlapping region, while its original predictions are retained in the non-overlapping region. In this study, the parameter r is defined to evaluate the reliability of the prediction of each pixel after multiscale fusion. Specifically, for a pixel in the non-overlapping region, we set $r = 1$. For a pixel in the overlapping region, if the prediction classes of each scale of images are the same for the pixel, we consider that the prediction is more reliable and set $r = 1.1$. Conversely, if they are not the same, we consider that the prediction is less reliable and set $r = 0.95$. For example, if three multiscale images overlap at a pixel, two multiscale images predict a building class, and the other multiscale image predicts a road; the prediction on this pixel is considered to be less reliable. Finally, the prediction results after multiscale fusion and the parameter r are used as the basis for selecting pseudo-labels in self-training.

3) Sample-Based Mobile Threshold: Previous methods typically require multiple recalculations of predicted masks as the threshold for pseudo-labels, in which a multiround training strategy is often utilized, thus inducing inconvenience for expert intervention. In this study, an end-to-end training process is adopted, which requires the dynamic adjustment and real-time updating of the threshold for different classes selected as pseudo-labels. In the case of N training samples and C classes in the target domain, this method only requires an $N \times C$ 2-D matrix to store the prior probability values, without causing memory burdens. This matrix is updated at each iteration and serves as the basis for subsequent threshold selection. Due to class imbalance, the threshold for selecting pseudo-labels should vary among different classes based on their predicted results. For the main classes, such as buildings, their threshold should be static; while for long-tailed classes, such as cars, their threshold should be determined based on the proportion of predicted labels after multiscale fusion, where a smaller proportion corresponds to a smaller threshold for them to be selected as pseudo-labels.

Specifically, for each softmax prediction of the student network, the prior estimation probability of pixels belonging to class $c \in C$ in sample $n \in N$ can be calculated as follows:

$$\chi_{c,n} = \frac{1}{h \times w} \sum_{i,j} m_{c,n,i,j} \tag{1}$$

where $m_{c,n,i,j}$ represents the softmax value of the pixel in row i and column j in sample n predicted to be class c; h and w represent the sizes after multiscale fusion. To ensure that the probability of each class slowly changes in the iterative process, the prior is updated by an exponential mobile average after each iteration t, which can be expressed as follows:

$$\chi_{c,n}^{t+1} = \gamma_\chi \chi_{c,n}^{t} + (1 - \gamma_\chi)\chi_{c,n} \tag{2}$$

where $\gamma_\chi \in [0, 1]$. For long-tail classes, we consider assigning them a lower threshold to make them easier to become pseudo-labels, which can utilize more pixels of these classes for supervision. The mobile threshold $\theta_{c,n}$ based on prior knowledge is defined as follows:

$$\theta_{c,n} = \zeta \left(1 - e^{\frac{-\chi_{c,n}}{\beta}}\right) m_{c,n}^{*} \tag{3}$$

where β and ζ are hyperparameters, as shown in Fig. 4, which together determine the threshold for different classes as pseudo-labels. $m_{c,n}^{*}$ represents the peak confidence score for category c in sample n

$$\mathbf{m}_{c,n}^{*} = \max_{i,j} m_{c,n,i,j}. \tag{4}$$

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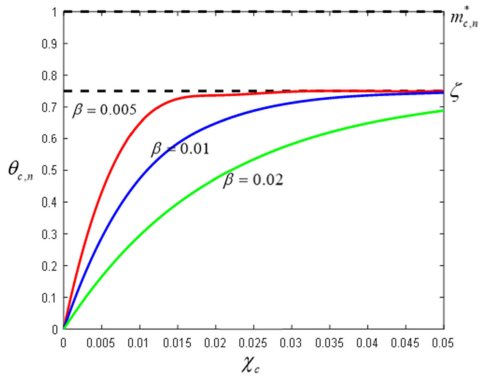


Fig. 4. Sample-based mobile threshold. Hyperparameters β and ζ are used to adjust the threshold selection for different classes. For the dominant classes such as impervious surfaces and buildings, $\chi_c \gg 0$, and their threshold is approximately $\zeta m^*_{c,n}$. However, for long-tailed classes such as clutter, $\chi_c \approx 0$, and its threshold is further controlled by β, with smaller χ_c leading to a smaller set threshold.

After the dominant class of the pixel is identified through $c^*_{n,i,j} = \max_c m_{c,n,i,j}$, the softmax prediction value of the dominant class on this pixel is $m_{c^*,n,i,j}$. The reliability index r is used for weighting and comparisons with the threshold $\theta_{c^*,n}$. The predicted class of pixels above the threshold is considered to be a reliable pseudo-label, while those below the threshold and nondominant class predictions will be ignored in the self-supervision loss

$$\mathbf{a}^*_{n,i,j} = \begin{cases} c^*_{n,i,j}, & r_{n,i,j} \times m_{c^*,n,i,j} > \theta_{c^*,n} \\ \text{ignore}, & \text{else}. \end{cases} \tag{5}$$

D. Training

1) Loss Function: For the loss of the source domain, since there is ground truth available, the cross-entropy loss function can be directly adopted, which can be expressed as follows:

$$\mathcal{L}^t_{n_{\text{sou}}}(p \mid \phi) = -y_{c,n_{\text{sou}}} \log(p_{c,n_{\text{sou}}}) \tag{6}$$

where p represents the predicted output with parameters ϕ; y represents the real label, and $n_{\text{sou}} \in N_{\text{sou}}$ represents the training sample from the source domain.

For the constraint of self-supervised generated pseudo-labels, the focus loss is used to handle sample class imbalance, which can be expressed as follows:

$$\mathcal{L}^t_{n_{\text{tar}}}(\overline{m}, m \mid \phi) = -m_{a^*,n_{\text{tar}}}(1 - \chi_{a^*})^{\lambda} \log(\overline{m}_{a^*,n_{\text{tar}}}) \tag{7}$$

where $\overline{m}$ represents the prediction of the teacher network with parameters ϕ; $n_{\text{tar}} \in N_{\text{tar}}$ represents the training sample from the target domain, and m represents the prediction of the student network after multiscale fusion. The parameter λ represents the focal point parameter, and a^* represents the pseudo-label of pixels. The probability of different pseudo-labels for classification is represented by χ_{a^*}, with lower values indicating long-tailed classes and higher $(1 - \chi_{a^*})^{\lambda}$ values indicating higher weights. Furthermore, we regularize the loss based on the confidence level $m_{a^*,n}$ of pseudo-labels, with higher confidence levels leading to higher weights for pseudo-label predictions.

For the alternately input source domain image n^{sou} and target domain image n^{tar}, they can be weighted by the total loss function, which can be expressed as follows:

$$\mathcal{L}^t_n = \mathcal{L}^t_{n_{\text{sou}}} + k\mathcal{L}^t_{n_{\text{tar}}}. \tag{8}$$

2) Training Process: The parameter update scheme is elucidated as follows. The teacher network uses the conventional SGD optimizer to update the network parameters; the student network performs momentum updates in each iteration, which can be expressed as follows:

$$\psi_{t+1} = \gamma_\psi \psi_t + (1 - \gamma_\psi)\phi \tag{9}$$

where ψ represents the student network parameter, ϕ represents the teacher network parameter, and the hyperparameter γ_ψ is used to adjust the update speed.

a) Pre-training: The alternating inputs of source and target domain images are used in the pre-training (without utilizing the labels from the target domain). The former updates the network parameters through backward propagation using the cross-entropy loss function. The latter only updates the mean and standard deviation (SD) in the batch normalization BN layers and freezes the rest of the model parameters. This avoids the recalculation of the mean and SD after pre-training.

b) Joint training: The pre-trained segmentation network is used as the teacher network, and its copy is used as the student network. For each batch input, the joint training steps are as follows:

1) Perform data augmentation on the input image by multiscale cropping and flipping.
2) For the source domain image, after the addition of photometric noise, brightness adjustment, and masking, input the source domain image to the teacher network. Use the cross-entropy loss function to guide training and perform the backward propagation to update the parameters of the teacher network.
3) Update the parameters of the student model with momentum.
4) For the target domain image, obtain the prediction mask through forward propagation and multiscale fusion of the student network, update the threshold with the mobile threshold method, and select reliable pseudo-labels.
5) After the addition of photometric noise, brightness adjustment, and masking, input the target domain image into the teacher network. Use the pseudo-label-based focus loss to guide training and perform the backward propagation to update the parameters of the teacher network.
6) Update the parameters of the student model with momentum.

The algorithmic flow of DA&MTSS can be summarized as Algorithm 1.

IV. Experiments

In this section, we present a comprehensive analysis of our proposed method, including detailed experimental results. Firstly, we provide a thorough description of our experimental setup in Section IV-A, encompassing dataset and task

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Algorithm 1 Algorithm Flow of DA&MTSS.

Input: Source domain data sample N_{sou} and label sample L_s for training, target domain data sample N_{tar};
Output: The teacher network parameter ϕ;

Pre-training of segmented networks

for $epoch = 1, 2, \ldots, max$ **do**
  for t in all batches **do**
    if sample $n_{sou} \in N_{sou}$ **then**
      Train f by $\min_{\phi}(L^t_{n_{sou}})$
    else if sample $n_{tar} \in N_{tar}$ **then**
      Train f with only update BN layer parameters
    end if
  end for
end for

Self-supervised training

Use pre-trained f as the initialization for teacher model $f_{tea}(\phi)$ and student model $f_{stu}(\psi)$.

for $epoch = 1, 2, \ldots, max$ **do**
  for t in all batches **do**
    if sample $n_{sou} \in N_{sou}$ **then**
      Generate $n^2_{sou} = \rho_2(\rho_1(n_{sou}))$
      Train f_{tea} by $\min_{\phi}(L^t_{n^2_{sou}})$
    else if sample $n_{tar} \in N_{tar}$ **then**
      Generate $n^1_{tar} = \rho_1(n_{tar})$
      Predict $m^1_{tar} = f_{stu}(n^1_{tar})$
      Multi-scale fusion $m_{tar} = \rho_3(m^1_{tar})$
      Identify reliable pseudo labels $m_0 = \vartheta(m_{tar})$
      Generate $n^2_{tar} = \rho_2(n^1_{tar})$
      Train f_{tea} by $\min_{\phi}(L^t_{n^2_{tar}})$
      Update φ of $M_{stu}(\varphi)$
    end if
  end for
end for

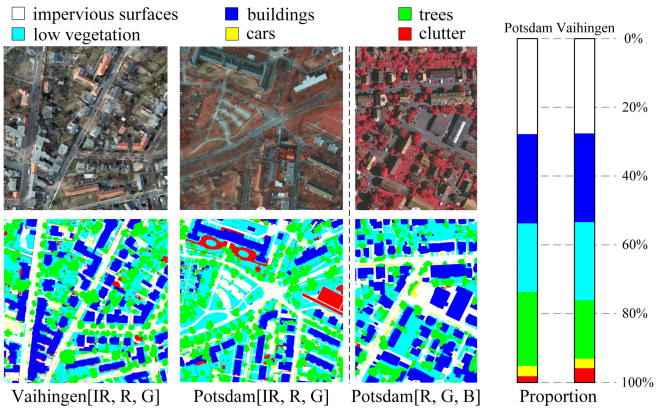


Fig. 5. Visualization and class distribution of pixels in the ISPRS Vaihingen and Potsdam datasets.

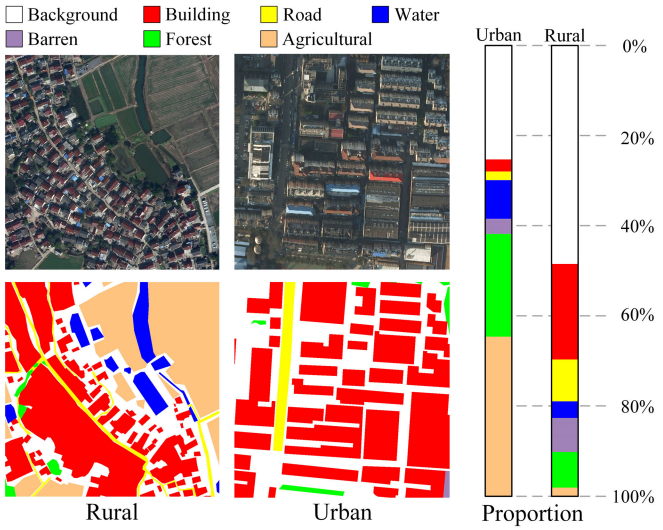


Fig. 6. Visualization and class distribution of pixels in the LoveDA datasets.

descriptions, parameter settings, and evaluation metrics. Subsequently, we present comparative experimental results of our proposed method in Section IV-B, demonstrating its superior effectiveness compared with existing methods.

A. Experimental Setup

1) Dataset and Task: The DA&MTSS is evaluated based on three publicly available datasets: ISPRS Vaihingen, Potsdam, and LoveDA. The datasets and task settings are elucidated as follows.

The Vaihingen and Potsdam datasets include semantic labels for all images, covering six common land cover classes. The Vaihingen dataset represents a relatively small village with sparsely distributed single and small multistory buildings, with a spatial resolution of 9 cm. The Potsdam dataset represents a historically significant city with more complex structures and streets, featuring a spatial resolution of 5 cm. Fig. 5 illustrates different scenes and the pixel distribution of each class in the training set. Three UDA semantic segmentation tasks are performed, namely: 1) P2V_S task, where the source domain is the Potsdam dataset with the channel composition [IR, R, G], and the target domain is the Vaihingen dataset with the channel composition [IR, R, G]; 2) V2P_S task, where the source domain is the Vaihingen dataset with the channel composition [IR, R, G], and the target domain is the Potsdam dataset with the channel composition [IR, R, G]; and 3) P2V_D task, where the source domain is the Potsdam dataset with the channel composition [R, G, B], and the target domain is the Vaihingen dataset with the channel composition [IR, R, G]. For each task, only the semantic labels of the source domain images are used, and several 256 × 256 patches are generated through mirror padding and cropping for each image.

The LoveDA [47] dataset includes two scenes from rural and urban areas of three cities: Nanjing, Changzhou, and Wuhan, China. The dataset comprises 5987 RGB images with a size of 1024 × 1024 pixels. All pixels are assigned to one of eight labels (including unlabeled). In the training process, the unlabeled region is ignored, and only seven classes are considered. Fig. 6 displays different scenes and the pixel distribution of each class in the training set. Following official guidelines, two UDA semantic segmentation tasks are defined: 1) R_U task (the source domain is the rural area, and the target domain is the urban area) and 2) U_R task (the source domain is the urban area, and the target domain is the rural area).

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TABLE I
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT UDA METHODS ON THE P2V_S TASK (%)

Postdam[IR,R,G]→Vaihingen[IR,R,G]									
	model	Source-only	MCD_DA [48]	AdaptSegNet [26]	FADA [49]	SDCA [50]	SPGAN-DA [38]	S&GDA [37]	Ours
IoU	Impervious surfaces	35.84	49.41	54.39	62.10	60.22	65.33	**78.01**	76.91
	Buildings	51.59	62.30	63.14	72.43	73.80	83.11	**87.42**	82.31
	Low vegetation	17.88	38.87	28.98	45.82	36.69	45.20	51.10	**66.08**
	Trees	54.59	57.66	52.65	61.74	62.86	66.04	70.20	**75.58**
	Cars	20.27	34.34	6.40	43.55	47.67	48.25	**58.75**	57.19
	Clutter	5.71	29.66	4.60	24.94	36.06	11.94	33.31	**40.55**
	mIoU	31.04	45.38	35.02	51.76	52.88	53.31	63.13	**66.43**

Each image is cropped into 256 × 256 patches for training purposes.

2) Parameter Settings: In this study, the method is implemented using the PyTorch framework on an RTX 3080 GPU with 12GB of memory. The segmentation network uses the DeepLab-V2 architecture with a ResNet-101 backbone. The SGD is used for optimization with a constant learning rate of 2.5×10^{-4}, a momentum of 0.9, and a weight decay of 5×10^{-4}. The batch size is set to 16, including two source domain images and two target domain images, with each image generating four images (one original image and three multiscale cropped images). The iteration number is set to 200. Next, the hyperparameters used in our method will be introduced. For data augmentation, the multiscale cropping area ranges from 0.5 to 1 times the size of the original image, and the flipping probability is 0.3. In the photometric noise, the Gaussian blur kernel radius is uniformly sampled within the range of [0.1, 2.0], and the color jitter is randomly and uniformly sampled within the range of [0.9, 1.1]. Brightness adjustment is randomly applied 1–3 times per image, with a weakening proportion ranging from 20% to 60%, and the random area size ranges from 0.125 to 0.25 times the size of the original image. Mask processing is randomly applied 0–2 times per image, with the random area size ranging from 0.0625 to 0.125 times the size of the original image. For pseudo-label generation, $\gamma_\chi = 0.99$, $\beta = 0.001$, and $\zeta = 0.75$. For the loss function, $\lambda = 3$ and $k = 5$.

3) Evaluation Metrics: The evaluation metrics that can best represent semantic segmentation performance, including IoU (per class) and mIoU (overall), are used to evaluate the effectiveness of our method. The formulae can be expressed as follows:

$$\mathrm{IoU}_i = \frac{\mathrm{TP}_i}{\mathrm{TP}_i + \mathrm{FP}_i + \mathrm{FN}_i}, \quad \mathrm{mIoU} = \frac{\sum_{i=1}^{M} \frac{\mathrm{TP}_i}{\mathrm{TP}_i + \mathrm{FP}_i + \mathrm{FN}_i}}{M} \tag{10}$$

where TP represents the number of correct predictions in this class; FP represents the number of correct predictions in this class; FN represents the number of predictions of this class into other classes; M represents the total number of classes. The higher these evaluation metrics, the better the performance of semantic segmentation.

B. Experiment Analysis

1) Comparative Experiment: The comparative experiments are conducted between models trained directly on the source domain (source-only) and several advanced UDA semantic segmentation methods in five task settings.

a) P2V_S task: The domain shift is mainly reflected in the spatial resolution, lighting and shadows, and geographic location. The quantitative experimental results are listed in Table I.

The source-only model, being affected by domain shifts, exhibits the worst performance, whereas all compared UDA methods demonstrate varying degrees of performance improvements. Our method achieves the best performance with an mIoU score of 66.43%. This can be attributed to the fact that we consider various reasons for domain shifts in RS images and conduct targeted data augmentation, thus contributing to a more robust segmentation network in cross-domain inference. In terms of per-class accuracy, the accuracy improvement for tail classes is more significant than that for major classes. For example, compared with the SDCA method, the mIoU score for the least represented clutter class increases by 4.49%, while the performance for more represented classes such as impervious surface and building is slightly lower compared with S&GDA. This is due to the fact that our method increases the contribution of long-tailed classes, thus avoiding the influence of class imbalance. The mobile threshold method dynamically lowers the threshold for selecting few-sample classes as pseudo-labels. Meanwhile, the focal loss based on confidence regularization assigns more weight to few-sample classes according to their proportion, increasing their contribution to the gradient. Fig. 7 illustrates the qualitative results in the target domain. Our method exhibits the best visual performance, particularly in addressing problems related to semantic prediction errors and rough segmentation boundaries, especially for clutter, cars, and low-vegetation classes.

b) V2P_S task: Due to the larger range of shadows and more complex lighting conditions in the Potsdam dataset, it is thus more challenging to use it as the target domain. Table II presents the quantitative experimental results, while Fig. 8 illustrates the qualitative results.

DA&MTSS achieves the best performance, reaching an mIoU score of 63.53%. It is worth noting that, for the clutter class, our method shows a precision improvement of at least 12.79%. As shown in Fig. 8, our domain adaptation results have more accurate semantic information, especially in long-tailed classes such as debris and vehicles. Meanwhile, our prediction results are smoother than other methods.

c) P2V_D task: The domain shift in this task extends beyond P2V_S and encompasses more significant inter-domain

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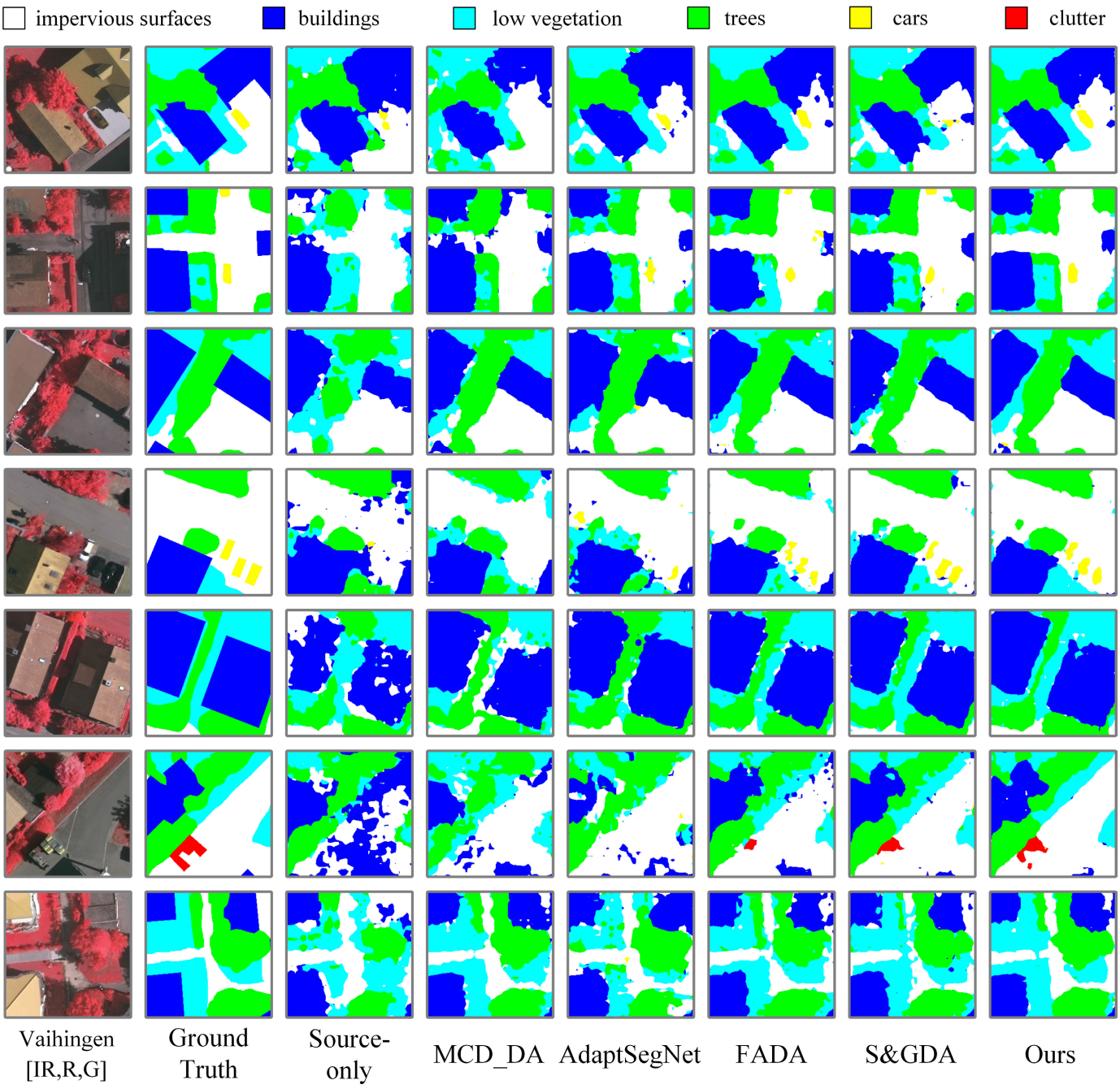


Fig. 7. Qualitative results of the P2V_S task.

TABLE II

QUANTITATIVE EVALUATION RESULTS OF DIFFERENT UDA METHODS ON THE V2P_S TASK (%)

	Vaihingen[IR,R,G]→Postdam[IR,R,G]							
	model	Source-only	MCD_DA [48]	AdaptSegNet [26]	FADA [49]	SDCA [50]	SPGAN-DA [38]	S&GDA [37]	Ours
IoU	Impervious surfaces	44.71	52.14	64.64	65.17	55.32	69.63	**76.52**	74.01
	Buildings	60.46	65.03	73.39	68.35	68.9	81.41	**84.28**	74.39
	Low vegetation	36.88	51.60	39.23	55.73	31.79	56.5	62.03	**63.18**
	Trees	63.46	60.39	62.90	17.02	57.96	57.34	66.03	**72.66**
	Cars	29.14	47.07	16.65	68.23	62.77	49.55	**72.51**	69.28
	Clutter	4.58	12.39	14.85	4.00	11.16	13.24	11.34	**27.64**
	mIoU	39.85	42.64	45.28	46.42	47.98	54.61	62.12	**63.53**

differences resulting from variations in imaging sensors. The quantitative results are presented in Table III, while the qualitative results are illustrated in Fig. 9.

DA&MTSS achieves the best performance, with a minimum increase of 1.77% in mIoU scores. This can be attributed to the fact that our method fully improves the contribution of

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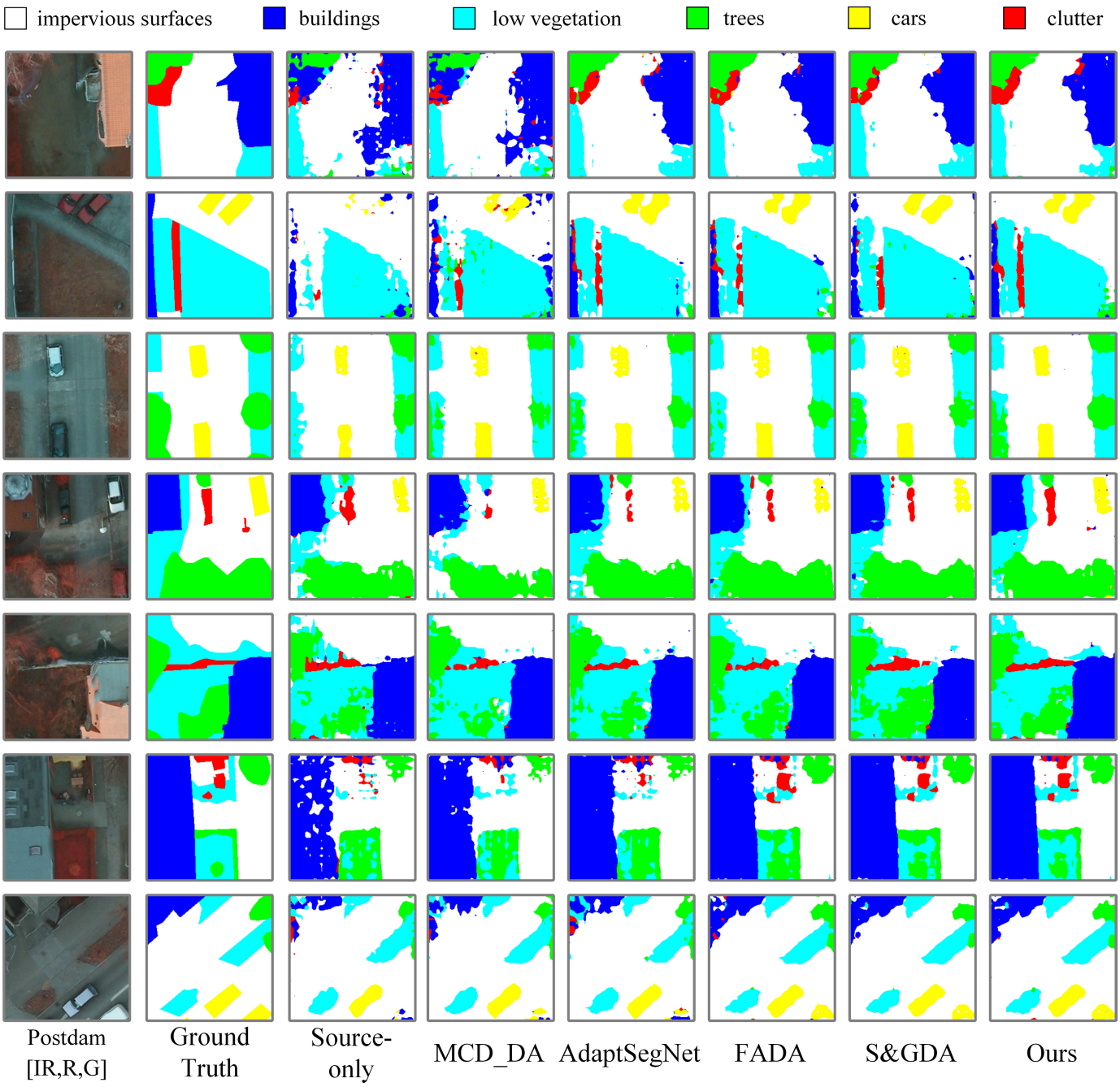


Fig. 8. Qualitative results of the V2P_S task.

TABLE III
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT UDA METHODS ON THE P2V_D TASK (%)

	Postdam[R,G,B]→Vaihingen[IR,R,G]								
	model	Source-only	MCD_DA [48]	AdaptSegNet [26]	FADA [49]	SDCA [50]	SPGAN-DA [38]	S&GDA [37]	Ours
IoU	Impervious surfaces	26.86	44.47	50.05	62.60	61.44	67.88	**68.93**	66.70
	Buildings	41.67	57.36	62.61	65.78	65.17	82.83	80.87	**83.25**
	Low vegetation	12.56	43.93	20.73	53.16	28.06	47.27	45.33	**49.35**
	Trees	44.48	52.72	56.45	14.45	54.23	**55.26**	50.29	52.27
	Cars	16.70	39.4	14.18	**65.66**	59.04	49.79	45.86	50.36
	Clutter	1.76	4.72	1.08	1.43	7.43	23.45	23.50	**35.17**
	mIoU	24.01	34.97	34.18	43.85	44.25	54.41	52.46	**56.18**

long-tail classes and significantly improves the single-class segmentation performance of low-value objects, cars, and debris classes. As shown in Fig. 9, the prediction results of the pure source model have large-scale semantic errors, indicating

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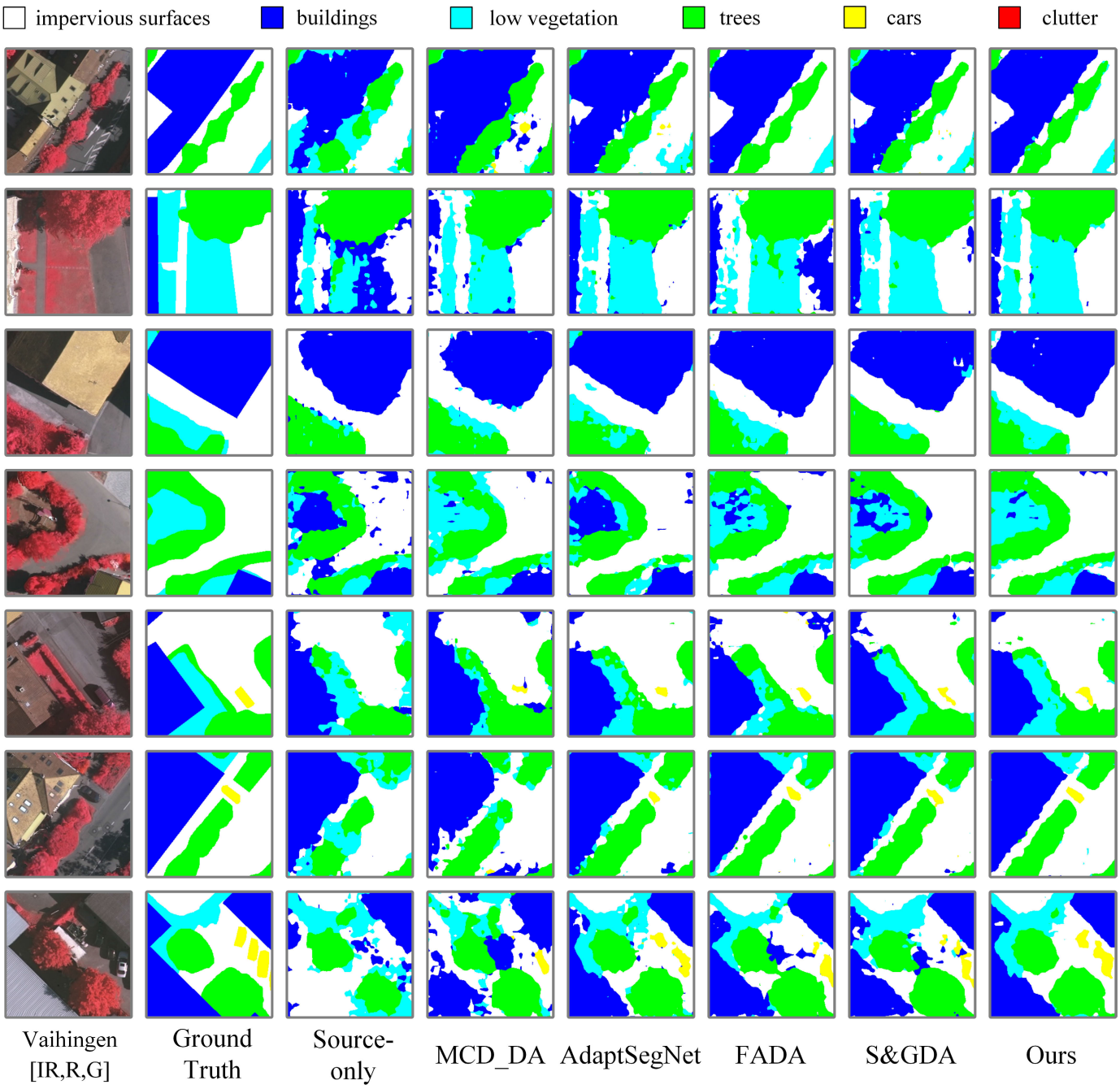


Fig. 9. Qualitative results of the P2V_D task.

TABLE IV
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT UDA METHODS ON THE U_R TASK (%)

		Urban[R,G,B]→Rural[R,G,B]							
	model	Source-only	MCD_DA [48]	AdaptSegNet [26]	FADA [49]	SDCA [50]	SPGAN-DA [38]	S&GDA [37]	Ours
IoU	Background	24.16	27.64	26.89	24.39	39.73	55.06	55.54	**58.81**
	Building	37.02	38.75	40.53	32.97	41.84	50.71	58.31	**61.68**
	Road	32.56	30.13	30.65	25.61	29.70	33.80	45.54	**47.37**
	Water	49.42	50.84	50.09	47.59	56.3	65.01	71.39	**73.93**
	Barren	14.00	18.15	16.97	15.34	12.21	9.07	14.39	**16.02**
	Forest	29.34	**35.43**	32.51	34.35	29.89	25.43	31.38	30.86
	Agricultural	35.65	26.27	28.25	20.29	37.87	55.44	63.13	**66.86**
	mIoU	31.74	32.46	32.27	28.65	35.36	42.07	48.53	**50.79**

a significant gap between domains. All four compared UDA methods not only display certain performance improvements but also exhibit serious semantic confusion, rough boundaries, and the phenomenon of long-tail classes being covered by the

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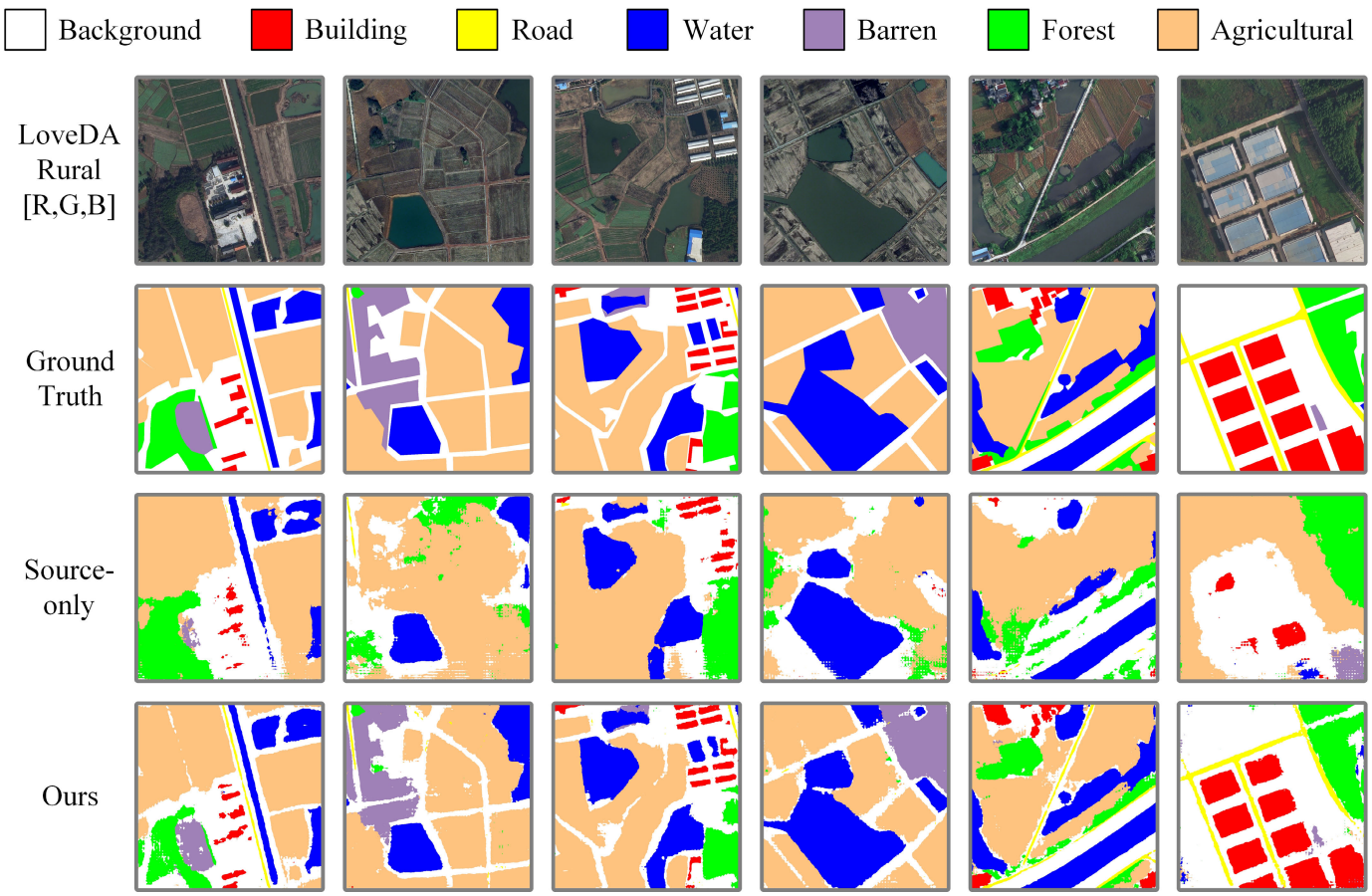


Fig. 10. Qualitative results of the U_R task.

TABLE V
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT UDA METHODS ON THE R_U TASK (%)

	Rural[R,G,B]→Urban[R,G,B]								
	model	Source-only	MCD_DA [48]	AdaptSegNet [26]	FADA [49]	SDCA [50]	SPGAN-DA [38]	S&GDA [37]	Ours
IoU	Background	43.30	**44.82**	42.35	43.89	41.63	35.45	40.04	40.66
	Building	25.63	26.67	23.73	12.62	28.33	50.31	**60.11**	59.32
	Road	12.70	16.16	15.61	12.76	23.89	51.02	58.68	**60.96**
	Water	76.22	81.08	**81.95**	80.37	77.95	68.42	74.03	79.38
	Barren	12.52	15.07	13.62	12.70	21.01	42.68	45.54	**48.31**
	Forest	23.34	28.03	28.70	32.76	32.41	40.17	48.93	**54.41**
	Agricultural	25.14	25.60	22.05	24.79	31.08	51.90	55.07	**59.50**
	mIoU	31.27	33.98	32.57	31.41	36.61	48.56	54.62	**57.51**

main classes. In contrast, our method has well-predicted long-tail classes while maintaining the segmentation performance of the main classes.

d) U_R task: The LoveDA dataset exhibits a severe class imbalance problem, and the disparity in class distribution between rural and urban areas is also significant. For example, the Farmland class occupies over 30% in rural areas, while it is only about 2% in urban areas. This difference highlights the importance of generating reliable pseudo-labels using the limited amount of Farmland class labels available in urban areas in the U_R task. Table IV provides the quantitative experimental results, while Fig. 10 illustrates the qualitative results.

DA&MTSS achieves the best performance, with a minimum increase of 2.26% in mIoU scores, and it also achieves the optimal per-class accuracy except for the Forest class. Our method effectively decreases the threshold for tail classes to be used as pseudo-labels, making classes similar to Farmland in urban scenes more likely to become self-training pseudo-labels. The confidence regularization loss function constrains these tail classes more effectively, allowing them to more effectively participate in training. As shown in Fig. 10, our method exhibits better accuracy compared with predictions from the pure source model, although some predictions still exhibit such problems as boundary fuzziness.

e) R_U task: Table V displays the quantitative experimental results, while Fig. 11 illustrates the qualitative results. In rural scenes, the proportions of buildings and roads are relatively low, presenting significant differences in appearance and features compared with urban scenes, posing more

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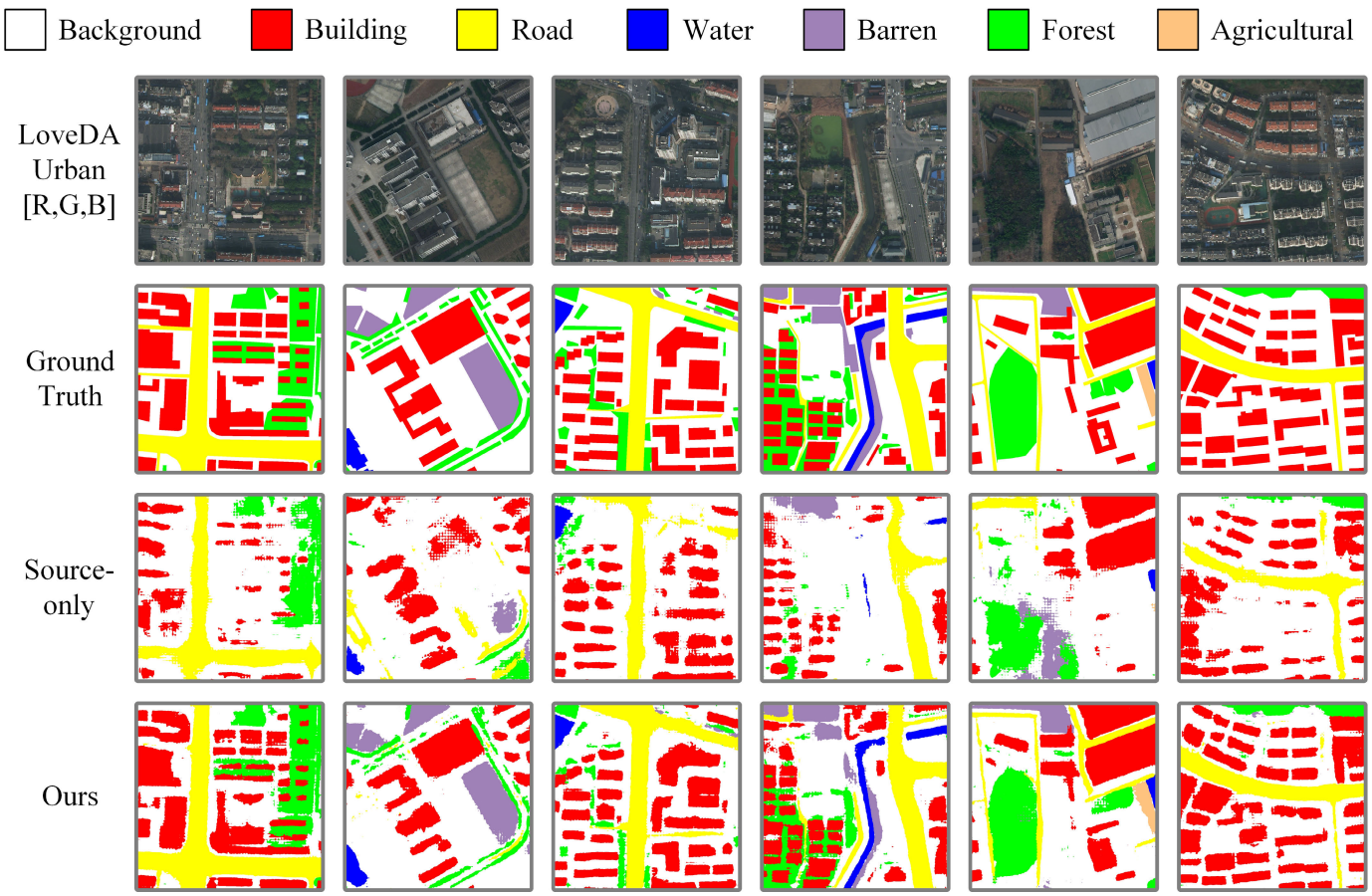


Fig. 11. Qualitative results of the R_U task.

TABLE VI
COMPARISON OF MIOU SCORES BETWEEN TRAINED TEACHER–STUDENT MODELS (%)

Model	Source Training	Source Test	Target Training	Target Test
Teacher	**74.65**	**74.04**	67.43	**66.43**
Student	71.33	70.60	**67.65**	64.52

challenges to the cross-domain inference for these classes. Our method achieves suboptimal and optimal performance in the building and road classes, respectively, significantly outperforming other methods. This result demonstrates the significant advantage of our method in handling tail classes and feature differences in cross-domain tasks.

2) Ablation Study: In this section, we discuss the comparison of teacher and student models after training is complete, explore the contribution of individual modules to experimental results through independent disabling, and investigate important hyperparameter settings (all conducted based on the P2V_S task).

a) Teacher–student model: Table VI shows the performance results of the trained teacher and student models in the source domain training set and test set, as well as the target domain training set and test set. The experimental results in the target domain test set can represent the performance of our method, which is consistent with our research significance. The experiment results demonstrate that the teacher model has significant performance advantages. In the design of our method, the teacher model takes alternately augmented source domain samples and target domain samples as the input, encouraging the teacher network to obtain better robustness. The student model takes "clean" target domain samples as the input and generates stable pseudo-label targets through slow momentum parameter updates. Therefore, the well-trained teacher and student models perform well in the target domain training set but exhibit significant differences in performance in unseen target domain test sets, with the teacher model showing better robustness and generalization ability.

b) Independent disabling: Table VII shows the results of ablation experiments conducted by independently disabling individual modules. It can be seen that multiscale cropping and fusion and momentum networks play important roles in our method, and their absence makes the network performance decrease by 8.18% and 6.80% in terms of mIoU scores, respectively. Multiscale cropping encourages the network to have better robustness to different spatial resolutions, while the average results of multiscale fusion can avoid the network's incorrect predictions to some extent. The momentum network can produce more stable pseudo-labels, even if the segmentation network undergoes significant changes in parameter updates. The parameters of the momentum network only have minor changes, outputting stable prediction results and providing stable pseudo-labels.

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TABLE VII
QUANTITATIVE EVALUATION RESULTS OF ABLATION EXPERIMENTS UNDER DIFFERENT CONFIGURATIONS (%)

	Configuration	mIoU	Δ
Data Augmentation	Disable multi-scale cropping and fusion	58.25	-8.18
	Disable horizontal flipping	65.82	-0.61
	Disable photometric noise	65.12	-1.31
	Disable brightness adjustment	65.59	-0.84
	Disable masking techniques	65.94	-0.49
Teacher-student model	Disable momentum updates(γ_ψ=0)	59.63	-6.80
Self-training	Disable class-specific thresholds(β=0)	63.56	-2.87
	Disable focal loss(λ=0)	63.54	-2.89
	Disable confidence regularization	65.10	-1.33
	Disable reliability parameters(r=1)	66.17	-0.26
Full framework		66.43	0.00

TABLE VIII
QUANTITATIVE RESULTS FOR MIOU UNDER DIFFERENT VALUES OF β AND ζ (%)

Parameter		ζ						
		0.5	0.6	0.7	0.75	0.8	0.9	0.95
β	0.0001	55.06	64.02	65.60	66.27	65.74	61.81	59.12
	0.001	55.63	64.16	66.08	**66.43**	66.38	62.64	59.30
	0.01	56.53	62.67	64.99	65.35	65.05	63.24	60.28

TABLE IX
QUANTITATIVE RESULTS FOR MIOU UNDER DIFFERENT VALUES OF γ_ψ AND γ_χ (%)

Paramater	0.5	0.8	0.9	0.99	0.999
γ_ψ	61.35	64.10	63.61	**66.43**	65.76
γ_χ	62.59	65.88	66.31	**66.43**	66.37

Other data augmentation techniques are also crucial for maintaining the advanced performance of the method. Disabling horizontal flipping (0.61%), photometric noise (1.31%), brightness adjustment (0.84%), and mask techniques (0.49%) all impair the training to some extent. Horizontal flipping simulates different imaging angles of sensors; photometric noise simulates different colors and textures between domains; and brightness adjustment and mask techniques are used to simulate different shadow intensities. These data augmentation techniques aim to weaken the imaging differences between domains and improve the robustness of the segmentation network.

If the threshold for all classes as pseudo-labels is set the same, mIoU would decrease by 2.87%. This means that long-tailed classes are difficult to become reliable pseudo-labels, and their segmentation performance will be seriously degraded. Similarly, without giving greater weight to long-tailed classes after disabling the focal loss, there would be a significant loss in accuracy (2.89%). Utilizing confidence regularization focal loss makes the accuracy increase by 1.33%, aiming to give higher weights to higher-confidence pseudo-labels.

c) Key hyperparameters: The important hyperparameter settings in the sample-based mobile threshold method and the loss function are also discussed.

Hyperparameters β and ζ jointly adjust the pseudo-label thresholds for different classes, as shown in Fig. 4. Table VIII demonstrates the quantitative results of different settings. It can be seen that a larger β value will significantly decrease accuracy, which can be attributed to the provision of excessively low confidence thresholds to long-tailed classes. Under this circumstance, the network is overly tolerant of incorrectly predicted long-tailed classes and treats them as reliable pseudo-labels. The parameter ζ seems to have slight impacts on performance, but this only exists in a local range between 0.7 and 0.8. If it is set to excessively large (larger than 0.9) or excessively small (smaller than 0.6), the prediction results will be either too difficult or too easy to become reliable pseudo-labels. This will lead to severe performance degradation.

The momentum coefficient γ_ψ governs the degree to which the student model preserves updates from the teacher model. By adjusting the magnitude of momentum, the update speed and direction can be influenced. Besides, the momentum coefficient γ_χ determines the rate at which the prior probability of each class evolves in the iterative process. These two coefficients jointly affect the stability of pseudo-label generation. Table IX presents the quantitative results for various γ_ψ values under the optimal value of γ_χ or vice versa. Empirically, both coefficients are set to 0.99 to achieve peak performance. This setting ensures that the student model retains a significant portion of its previous parameters during each parameter update, with only minor adjustments stemming from the updated teacher model. Meanwhile, the prior probability of each class evolves gradually throughout the iterative process.

When γ_ψ is set to a higher value, the parameter updates of the student model become overly reliant on the current gradient, leading to premature and suboptimal convergence. Conversely, a smaller value of γ_ψ induces the risk of oscillation, resulting in unstable updates of the student model's parameters and consequently unstable pseudo-labels. These inaccurate pseudo-labels further complicate the problem by affecting the teacher model's parameter updates, thus leading to severe performance degradation. In contrast, the influence of γ_χ is relatively minor. This is due to the fact that, when γ_ψ is set to its optimal value, the student model's predictions exhibit robust stability. Since the prior probability of each class inherently varies little throughout the iterative process, the momentum update further minimizes these variations.

The focus parameter λ is used to adjust the weight of different classes in the loss calculation. To make the network pay more attention to the long-tail classes that are difficult to segment during training, it is necessary to appropriately

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TABLE X
QUANTITATIVE RESULTS FOR MIOU UNDER DIFFERENT VALUES OF λ (%)

Paramater	1	2	3	4
λ	65.75	66.30	**66.43**	66.14

TABLE XI
QUANTITATIVE RESULTS FOR MIOU UNDER DIFFERENT VALUES OF k (%)

Paramater	0.1	1	3	5	10
k	64.88	65.87	65.60	**66.43**	64.24

increase their relative weight. Table X shows the quantitative results under different values of λ. An excessively small value of λ will still result in insufficient attention being paid to long-tail classes, while an excessively large value of λ will enable the network to ignore the main classes to some extent in the training process.

The parameter k is used to balance the cross-entropy loss in the source domain and the focus loss in the target domain. Besides, different values of k are also compared in this study, as shown in Table XI. After giving higher weight to the focus loss, the network is more inclined to correctly predict the target domain, which is consistent with the purpose of our method. However, an excessively high weight will ignore the true label constraints in the source domain, resulting in a decrease in the quality of generated pseudo-labels, which in turn reduces the network's prediction accuracy for the target domain.

V. CONCLUSION

Our study develops a new framework DA&MTSS to address the unique challenges of RS images in UDA semantic segmentation tasks. We first analyze the common factors that cause domain shifts in RS images and adopt various data augmentation techniques to attenuate domain shifts. In addition, to obtain stable pseudo-label targets, we design a sample-based mobile threshold method that can dynamically update the threshold of both the dominant and the long-tail classes as pseudo-labels during training. The end-to-end DA&MTSS can achieve the co-evolution of network parameters and pseudo-label quality, so only one round of training is required and no additional expert knowledge is needed. Comprehensive experiments conducted in five task settings on three datasets showed that DA&MTSS consistently achieved higher mIoU scores, demonstrating its performance advantage in semantic segmentation of RS images in UDA. The ablation experiment shows that the innovative module we designed has a significant boosting effect on segmentation performance. In future endeavors, we aim to investigate strategies for UDA in semantic segmentation tasks, particularly in datasets exhibiting diverse domain shift phenomena.

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